Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 15

Relations and Their Properties

Covers Chartrand & Zhang, Section 5.1

Status of this set

Ungraded practice. Solutions released with it. Attempt everything first.

Part I: How to Practice This Material

Prove with an argument, disprove with names

Each property is a universal statement

	To prove	To disprove
Reflexive	argue for arbitrary a	name one a with $a \not R\, a$
Symmetric	argue for arbitrary a, b	name a, b: $a\, R\, b$, $b \not R\, a$
Transitive	argue for arbitrary a, b, c	name a, b, c

A disproof of transitivity needs *three* named elements, and you must state that the first two links hold and the third fails.

Three mistakes to expect

1. **"It's not transitive" with no witnesses.** Name the elements.
2. **Assuming the properties come as a package.** They are independent.
3. **Skipping edge cases.** Zero, equal elements, and the empty set break more relations than anything else.

Part II: Practice Problems

Section A — Relations given as lists

1. Let $A = \{1, 2, 3\}$. For each relation, state which of reflexive, symmetric, transitive it has, with a witness for each failure.

(a) $R_1 = \{(1,1), (2,2), (3,3), (1,3), (3,1)\}$

(b) $R_2 = \{(1,2), (2,3), (1,3)\}$

(c) $R_3 = \{(1,1), (2,2), (3,3), (1,2), (2,3)\}$

(d) $R_4 = \emptyset$

2. Let $A = \{1, 2, 3, 4\}$. Give an example of a relation on A that is

(a) symmetric but not reflexive;

(b) reflexive and symmetric but not transitive;

(c) reflexive and transitive but not symmetric.

Section B — Relations on $\mathbb{Z}$ and $\mathbb{R}$

3. For each relation on $\mathbb{Z}$, determine which properties hold. Give a witness for each failure.

(a) $a \; R \; b$ if $a + b$ is odd.

(b) $a \; R \; b$ if a divides b.

(c) $a \; R \; b$ if $a - b$ is divisible by 3.

(d) $a \; R \; b$ if $|a - b| \leq 1$.

4. A relation is defined on $\mathbb{R}$ by $x \; R \; y$ if $x - y \geq 0$.

(a) Give two reals related by R and two that are not.

(b) Determine which properties R has, with witnesses.

Section C — Counting

5. Let $|A| = n$.

(a) How many relations are there on A?

(b) How many are reflexive?

(c) For $n = 3$, give both numbers.

Section D — Writing

6. In five to eight complete sentences, explain why disproving that a relation is transitive requires naming three elements, while disproving reflexivity requires only one. Illustrate each with a specific relation.

Section E — Challenge (optional)

7. Prove that if R is symmetric and transitive and *every element of A is related to at least one element*, then R is reflexive. Then explain why the extra hypothesis cannot be dropped.